Overall your text reads well. Be aware of text and even word redundancies and smooth transitions. Once you know what you have as „results" shift the text to presenting results only (in this aspect paper (results only) and theses & questions, methodology, results) differs.

LaTeX The Latency of Style: Assessing Stylistic Agency in Real-Time Diffusion for Live Audio-Visual Performance

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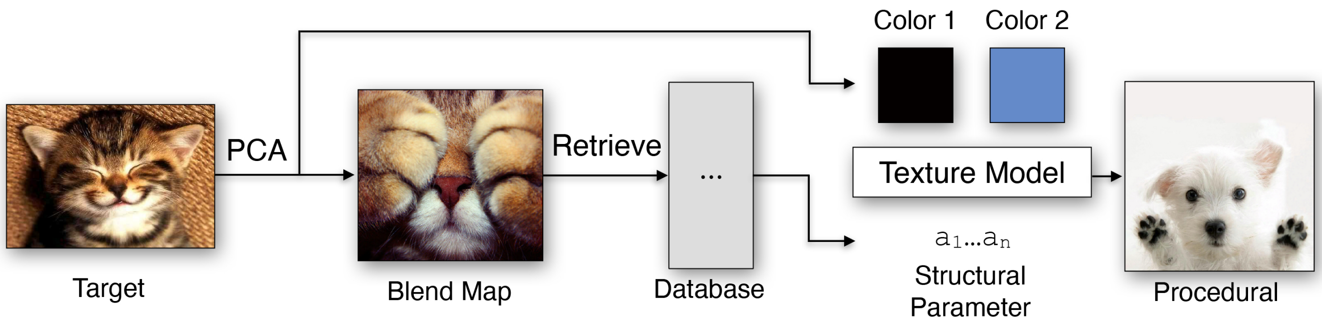

Figure 1: *Teaser Image. Explain the image.*

Do not use the same text in abstract and intro. The abstract can be more high-level – or „abstract" :)

Abstract

Real-time diffusion pipelines like StreamDiffusion are increasingly used by audio-visual performers, enabling live generation within environments like TouchDesigner. While StreamDiffusion supports interactive generation through text prompts and audio/video modulation, it remains unclear how much genuine stylistic agency artists have over the output. This is especially true because StreamDiffusion depends on baked TensorRT acceleration engines, which makes integrating custom Low-Rank Adaptations (LoRAs) difficult. This paper investigates to what extent artists can leverage StreamDiffusion's parameters and custom LoRAs within TouchDesigner to gain stylistic control without sacrificing real-time performance. I implemented a StreamDiffusion pipeline in TouchDesigner, evaluated existing control methods (text prompting, audio/video modulation), and (attempt to) incorporated a custom-trained LoRA. These are evaluated qualitatively, through observations on artistic control, and quantitatively, through performance metrics like latency and fps. (Results will follow)

1. Introduction

Recent advances in diffusion models expanded the opportunities and use cases for AI-generated visual content. ~~While early diffusion systems needed seconds or even minutes to create an image,~~ current models are even close to real-time generation. This advantage is especially interesting for artists in the field of audio-visual performances, where they slowly start to integrate these systems into their workflow. Maybe add an example here ?!

Transition?

TouchDesigner is a node-based visual scripting environment that is widely used for interactive live installations and performances. It allows for the integration of other software and data from hardware components and facilitates real-time rendering and audio-reactivity. The integration of diffusion models into this workflow additionally enables artists to blend their art with AI-generated content in real-time,* making visuals more versatile. This is possible through the open-source pipeline ~~called~~ StreamDiffusion. Instead of changing the actual AI models, StreamDiffusion changes how the data flows, using pipeline-level tactics to improve latency, fps, and visual stability.

You integrate a lot !!

The only? option.

Hard to understand

Despite those ~~new~~ developments, it remains difficult to gain full control over the generated output. Existing workflows primarily rely on standard text prompts and predefined model behavior which artists modulate externally using audio and video data. While this offers dynamic interactivity, it remains unclear to what extent this

?

* again, add an example for this? Paint a picture „of the scenario".

This is your main motivation and it needs more details! What is currently missing (example) and what do you add in terms of artistic control?

pipeline allows artists to include their individual handwriting. Furthermore, because StreamDiffusion relies heavily on baked TensorRT acceleration engines for real-time speed, integrating and swapping Low-Rank Adaptations (LoRAs) fluidly is highly limited, which further limits the visual guiding of the model.

A paper rather presents results not questions

For this reason the following research question is addressed: To what extent can artists leverage StreamDiffusion's parameters and custom LoRAs in TouchDesigner to assert stylistic agency without compromising real-time performance metrics?

To further close this gap...

To ~~answer this question~~, a Streamdiffusion pipeline is implemented into TouchDesigner to identify and evaluate existing control methods like text prompting and audio/video input modulation. A further attempt is to incorporate a custom-trained Low-Rank Adaptation (LoRA). These control methods are then evaluated through qualitative observations on artistic control and ~~also~~ quantitatively, through evaluation metrics, ~~measuring their effect on performance parameters~~ like latency and fps.

2. Related Work

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3. Main Content - Name and Structure the Main Content According to your Paper

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3.1. Method

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3.2. Results

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3.3. Discussion

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4. Formatting your paper

4.1. Bullet Points

The following is an example for a bullet points list. The contributions of this paper are

- my greate archivement number one,
- a second awsome insight,
- and last but not least the third contribution.

4.2. Figures

All graphics should be centered.

If your paper includes images, it is important that they are of sufficient resolution to be faithfully reproduced.

Figure 2: *Here is a sample figure. All figures must have sensible captions and must be referenced in the text.*

Instructions from the original conference template (for the ACS FUB it is not that relevant):

To determine the optimum size (width and height) of an image, measure the image's size as it appears in your document (in millimeters), and then multiply those two values by 12. The resulting values are the optimum x and y resolution, in pixels, of the image. Image quality will suffer if these guidelines are not followed.

Example 1: An image measures 50 mm by 75 mm when placed in a document. This image should have a resolution of no less than 600 pixels by 900 pixels in order to be reproduced faithfully.

This is an examplary reference to Figure 3, which shows the incredible test image from the Eurographics conference.

4.3. Tables

A simple table might look like the following. However there are several ways to create and layout tables - it is up to you to style a table up to your liking. This is a reference to Table 1.

4.4. Formulas

In LaTeX it is fairly easy to add mathematical symbols and formulas. See the following as example [wik19]:

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Table 1: *Here is a sample table. All tables must have sensible captions and must be referenced in the text.*

$$a = bq + r \tag{1}$$

which is true if a and b are integers with $b \neq c$.

$$L' = L\sqrt{1 - \frac{v^2}{c^2}} \tag{2}$$

4.5. Footnotes

Please do not use footnotes at all.

4.6. References

References are layouted automatically. Add the needed entry to the bibtex file and use the citation command to reference the entry.

If you don't want to mention the author's name just add the citation to the end of the sentence [Per85]. You can also cite more than one references at the same time [GSH18, Per85, SLKD15].

Sometimes you want to refer to the authors name. With this template you need to write out the names manually. For more than two authors you only use the name of the first author and abbrivate the remaining names with "et al."

An example for more than two authors:

Raul et al. [RRDR09] thoroughly review the different approaches for segmentation algorithms. Our problem requires the segmentation of color images, summarized by Cheng et al. [CJSW01].

An example for two authors:

Also aiming for a fragmented aesthetic appeal by transforming original image data "piecewise", Collomosse and Hall [CH03] render from salient image features a cubist version of an input. Similarly, Lai and Rosin [LR13] loosely aim for a cubist appearance by segmenting an input and by simplifying the segments into constant colors, while maintaining global structures.

An example for one author:

Artistically, we are inspired by the art of Wehrli [Weh03], who manually creates image decompositions and rearrangements.

5. Conclusion

Please direct any questions to the production editor in charge of these proceedings.

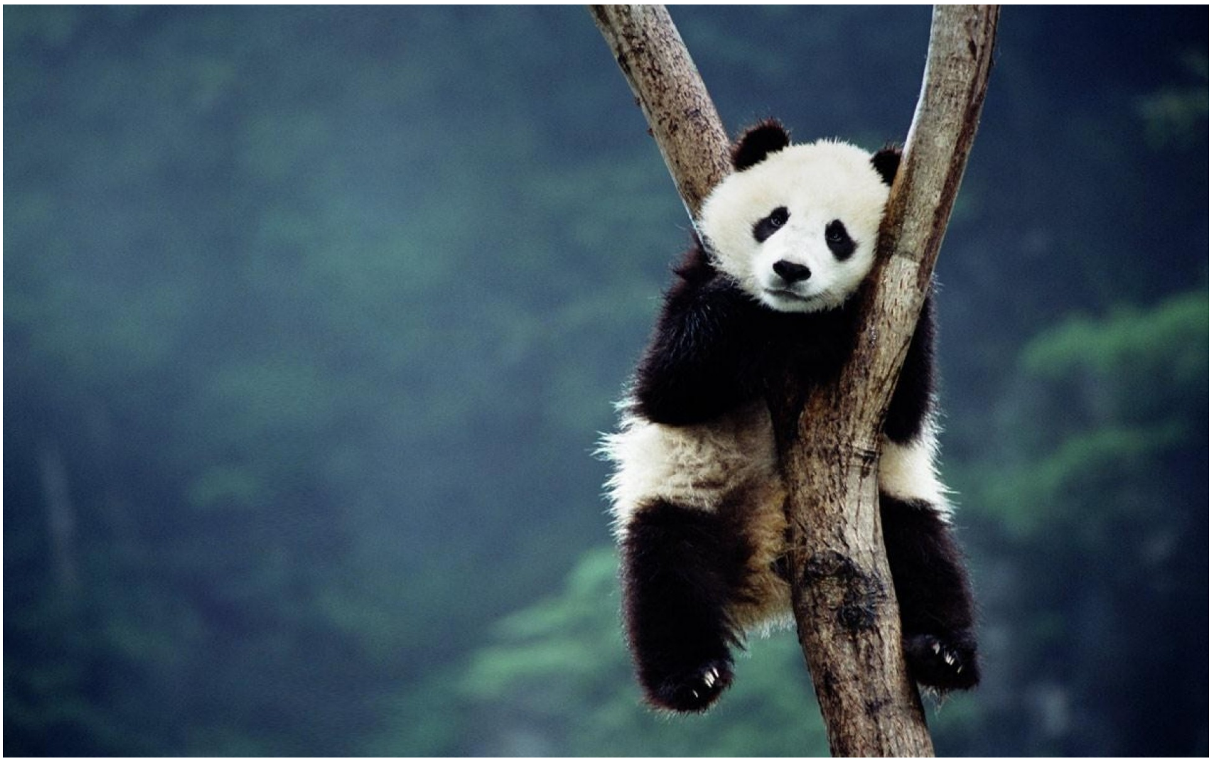

Figure 3: *Two-column width figure*

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